

Problem-Solution Fit

Date	28 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENTS [CS] - Students using online learning platforms - College and university students - Learners preparing for examinations - Self-paced online learners - Educators supporting student learning	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited understanding of difficult concepts - Lack of personalized guidance - Time constraints during study - Emotional stress while learning - Difficulty maintaining motivation	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: Google Search, ChatGPT, Gemini AI, YouTube Tutorials, LMS Pros: - Easily accessible - Large learning resources - Quick answers Cons: - Cannot detect learner emotions - Generic responses - Limited personalized learning support				
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty understanding complex topics (Very Frequent) - Lack of personalized learning support (Frequent) - Feeling confused or frustrated while studying (Frequent) - Reduced motivation during learning (Occasional) - Generic responses from learning platforms (Frequent)	7. BEHAVIOR + ITS INTENSITY [BE] - Searches online for explanations (High) - Watches educational videos (High) - Uses AI tools for learning (Medium) - Revises topics repeatedly (Medium) - Requests help from teachers or friends (Low)	8. CHANNELS OF BEHAVIOR [CH] <table border="0"> <tr> <td>ONLINE</td><td>OFFLINE</td></tr> <tr> <td> - Google Search - ChatGPT - Gemini AI - YouTube - Online Learning Platforms </td><td> - Classroom Learning - Faculty Guidance - Study Groups - Library - Peer Discussions </td></tr> </table>	ONLINE	OFFLINE	- Google Search - ChatGPT - Gemini AI - YouTube - Online Learning Platforms	- Classroom Learning - Faculty Guidance - Study Groups - Library - Peer Discussions
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3. TRIGGERS TO ACT [TR] - Preparing for examinations - Difficulty completing assignments - Encountering challenging concepts - Seeking personalized academic guidance - Looking for emotional support while learning	4. EMOTIONS BEFORE / AFTER [EM] Before: Confused, Frustrated, Stressed, Demotivated, Uncertain After: Confident, Motivated, Curious, Focused, Supported	9. PROBLEM ROOT / CAUSE [RC] - Learning platforms cannot understand student emotions. - Generic learning resources do not adapt to emotional state. - Lack of personalized academic assistance. - Difficulty identifying learners' emotional needs. - Insufficient motivation during difficult learning sessions.	10. YOUR SOLUTION [SL] Emotion Detection & Learning Support Engine - BiLSTM + BERT emotion detection - Google Gemini AI guidance - Mixed emotion detection - Personalized learning recommendations - Analytics dashboard and session history			